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Full Length Article

Effect of ZnO nanoparticles concentration as additives to the epoxidized *Euphorbia Lathyris* oil and their tribological characterization

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ABSTRACT

The disposal of the synthetic esters creates pollution and the depletion of the petroleum reserves is the issue of concern. The vegetable oil with application of nanoparticles are one of the feasible ways for providing an alternative. This study presents the tribological characteristics of *Euphorbia Lathyris* oil. In the initial phase, the oil was chemically modified through the epoxidation process and further addition of the zinc oxide (ZnO) nanoparticles was done. The nano lubricants were evaluated for the thermophysical characterization like viscosity, flash point, pour point with tribo analysis like friction behaviour, anti-wear mechanism, and worn surface examination using Scanning Electron Microscope. The test was performed by using a pin on disc tribometer for the tribological analysis. The viscosity of the oil shows increment with chemical modification and increase in addition to the nanoparticles. The maximum increment was observed with 8% at 100 °C temperature. The flash point increases with the addition of the nanoparticles but decrement in the pout point was observed with an increase in nanoparticles concentration. During the flow behaviour analysis, all the nano lubricant samples showed Newtonian behaviour by presenting a linear relationship between the shear rate and shear stress. During the investigation, 0.5% concentration of ZnO nanoparticles demonstrated a reduction in the coefficient of friction (COF) and wear of the pin of about 8.23% and 5.13% respectively. The SEM images also show better surfaces at 0.5% concentration due to effective lubrication to the surface.

1. Introduction

Lubrication is the process which assists in reducing friction by providing a defensive film between two mating parts through the lubricant application. The lubricity of the parts gets improved if the lubricant contains thermal stability, higher viscosity index, low freezing point, high boiling point, and capable to resist the oxidation process [1–3]. Several components of the engine require effective lubrication for proper functioning with minimum adverse effect on the environment as shown in Fig. 1. Improvement in the tribological performance of yields can lead to increase engine power output, reduced oil/lubricant consumption, improved durability, reliability and engine life, and reduced maintenance requirements and long service intervals.

Synthetic oils are one of the petroleum products which reduces friction between the parts during their motion and improved their life. But the use of synthetic oil and their disposal is one of the issues which need to manage as they are harmful to the environment. They produce toxic particles during their combustion and with their disposal aquatic environment gets polluted [4]. The use of mineral oil is enormous among oil-based commodities due to its application for lubrication reasons [5]. There is a need to find a substitute that could replace synthetic oil and provide effective lubrication. In keeping this point as a background of the study, the author concentrated on the exploration of bio-based lubricants [6]. Bio-based lubricant is one of the assets and plays a key role in the improvement of sustainability, which is progressively economic and harmless to nature during its usage [7]. Farfan-

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Cabrera *et al.* [3] assessed the effect of thermal ageing on the lubricity of the jatropha oil during its usage in engines. There was no effect on the frictional behavior observed by thermal ageing but the wear was reduced by its effect. Comparative lubricity of coconut and mustard oil was assessed by Sajeeb *et al.* [8]. They are blended with each other in different proportions. The 5% blend presented improved friction and wear behavior with comparison to the mineral oil. Hisham *et al.* [9] blended waste cooking oil with engine oil to investigate tribological behavior. The blends of 5% and 10% increase the coefficient of friction and wear due to abrasion on the surface.

Bio based lubricant has better biodegradability compared to oil-based synthetic oils, but certain limitations are there which need to be addressed for proper tribological applications [10]. Physicochemical characteristics were important for the use of substitute oils such as viscosity, flash point and fire point in addition to enhancing the physicochemical attributes, nanoparticles are one of the suitable additives which could enhance bio based lubricant properties. Peng *et al.* [11] evaluated the tribological characteristics of bio-oil. The solid superacid significantly promotes the catalytic reaction. The additives are required to boost the performance of the non-edible oils for the tribological applications. Anwar *et al.* [12] checked the performance of oxygenated additives to the papaya biodiesel and improvement in the performance of the diesel engines was observed.

In the previous studies, nanoparticles are added to boost the properties of conventional synthetic oils and only a few studies were conducted based on their usage to the bio-based lubricants [13–15]. Bhaumik *et al.* [14] developed an artificial neural network model for predicting anti-wear properties of the lubricant during the implementation of nano friction modifiers. Asnida *et al.* [16] added copper oxide to the engine lubricant. Improved surface wear was attained at 0.005% addition. Peng *et al.* [17] examined the effect of silicon dioxide nanoparticles size on the lubricant. An improvement in the tribological behavior was observed. Wang *et. al.* [18] expressed that the hexagonal boron nitride (hBN) nanoparticles are appropriate for the decrease of coefficient of friction and wear when they are added to the castor oil. The hBN nanoparticles addition showed less friction and wear of the parts. It is clear from the above studies that optimum concentration of the nanoparticles are suitable for proper enhancement of the lubricity of the bio based lubricants.

Fig. 2 shows the effect of nanoparticles concentration, the proper proportion of the nanoparticles is suitable for assisting lubricant film in a more effective way to support the pressure on the film. Non edible vegetable is suitable as a feedstock for the bio-based lubricants as they contain fatty acids having double bonds between their structure which could assist in resisting applied pressure on the lubricant films. The application of zinc oxide (ZnO) nanoparticles is suitable as an additive to the bio-based lubricants as it contains higher surface energy, proper dispersion in the oil and better thermal conductivity.

In the previous studies, most of the studies were performed while considering edible oil for the tribological application. The application of the non edible oil based lubricants was limited and the studies while considering nanoparticle application to the non-edible oil was limited. This study will be beneficial to the industries where research is focused on finding out bio-based lubricant as products. In the earlier studies, the application for tribological analysis of non-edible oil with nanoparticles as additives were not performed. It was only limited to the edible oil and engine oil. The application of zinc oxide nano-particles to *Euphorbia lathyris* oil will provide in-depth knowledge to the readers and industries involved in this field. These type of bio-based lubricants find their application as automotive fluid, hydraulic fluid, gear oil and fluid for machining [20]. *Euphorbia lathyris* tree belongs to the family - Euphorbiacae, can produce a significant amount of oil. The tree used to bear fruits every second year and grow in drought and arid soils. They are mostly found in the Eropoe, Southwest Asia, India and Northwest Africa. Around 1.5 to 2.5 tons per hectare per year seeds were produced from the tree and contains around 49% oil [21–23]. Based on its availability, there is a need to explore its suitability as a lubricant applicable for the tribological investigation.

In the present investigation, raw *Euphorbia Lathyris* oil has been considered as the reference oil, and further chemical medication through epoxidation process was performed to improve its oxidation stability. In addition, ZnO nanoparticles are added to the modified oil to

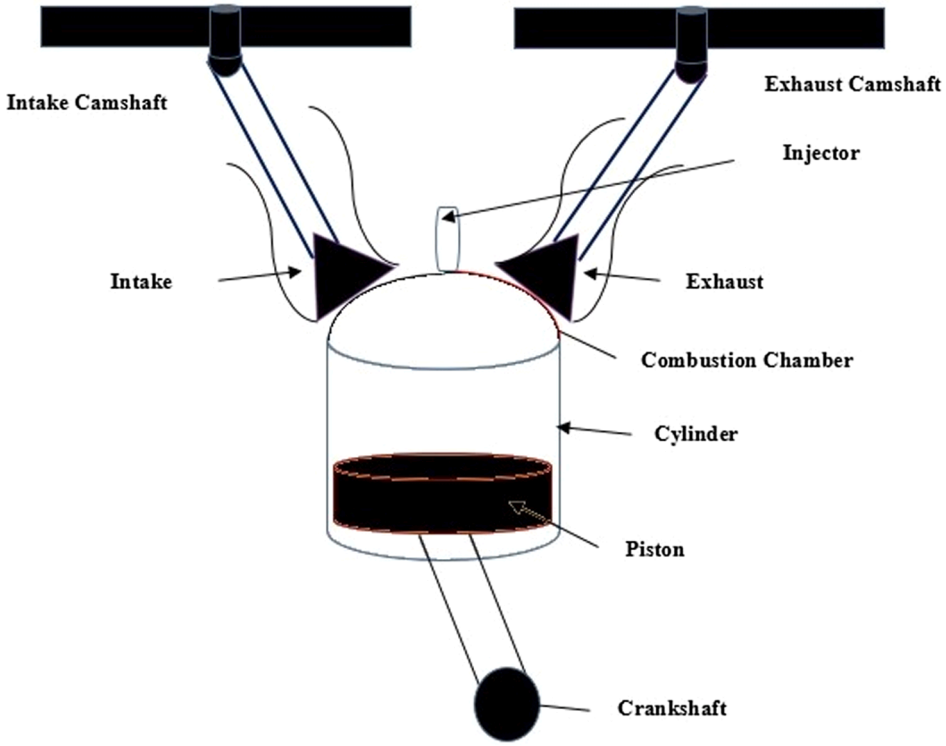


Fig. 1. Engine components showing lubricating parts.

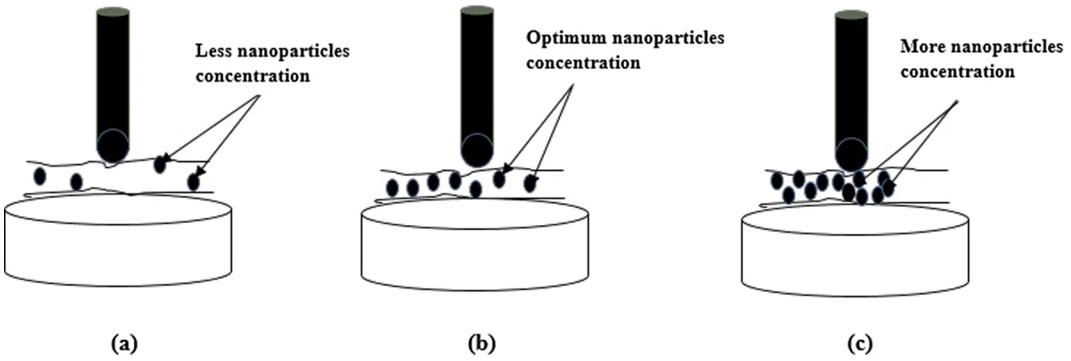


Fig. 2. Effect of nanoparticles concentration on the lubrication mechanism [19].

check its lubrication mechanism. The physicochemical properties of the modified oil with nanoparticles as additives were analyzed. For checking the stability of the nanoparticles to the modified lubricant, ultraviolet spectroscopy was used. The tribological characterization was performed using Pin on disc tribometer. A Scanning electron microscope (SEM) was used to examine the surface topography of the pin samples analyzed.

2. Materials and methods

The approach adopted in this study is shown in Fig. 3. This shows the process involved during this study. The detailed discussions about the process are ellaborated in further sections.Fig. 4.

2.1. Composition of Euphorbia Lathyris

For the evaluation purpose, raw *Euphorbia Lathyris* oil was procured from the M/s Pallishree Limited, Kolkata. In order to endorse the worldwide commercial viability of biodiesel, low-cost non-edible oils including *Euphorbia Lathyris* may be used as feed stocks and Euphorbia Lathyris is viewed to be a possible source of petroleum-like crops [23]. Since ZnO nanoparticles are environmentally sustainable, they are often used for different applications in tribology [24].

The oil mainly contains triglyceride esters of fatty acids and glycerol which were estimated according to European standard method EN14103:2003.

The fatty acid composition of *Euphorbia Lathyris* oil is analyzed by a Gas Chromatography-Mass Spectroscopy (GC-MS) (GC-10, Shimadzu, Japan) having a capillary column with 30 mm length and 0.25 mm diameter lined with a 0.25 μm (Rtx-5 ms, Rextex). Samples were injected in split/column flow ratio 24:1. Helium was used as the carrier gas with a flow rate of 1 ml/min. The injection temperature was maintained at 250 °C and column oven also 250 °C (programmed to start at 120 °C, held at this temperature for 5 min and heated at a rate of 3 °C/min to 250 °C).

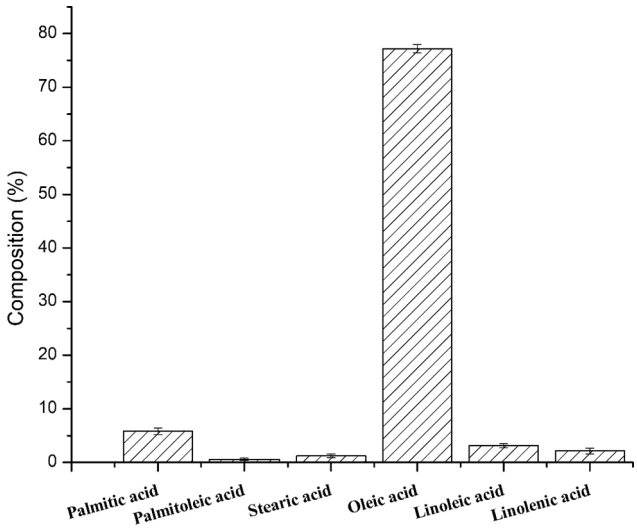


Fig. 4. *Euphorbia Lathyris* oil fatty acid compositions.

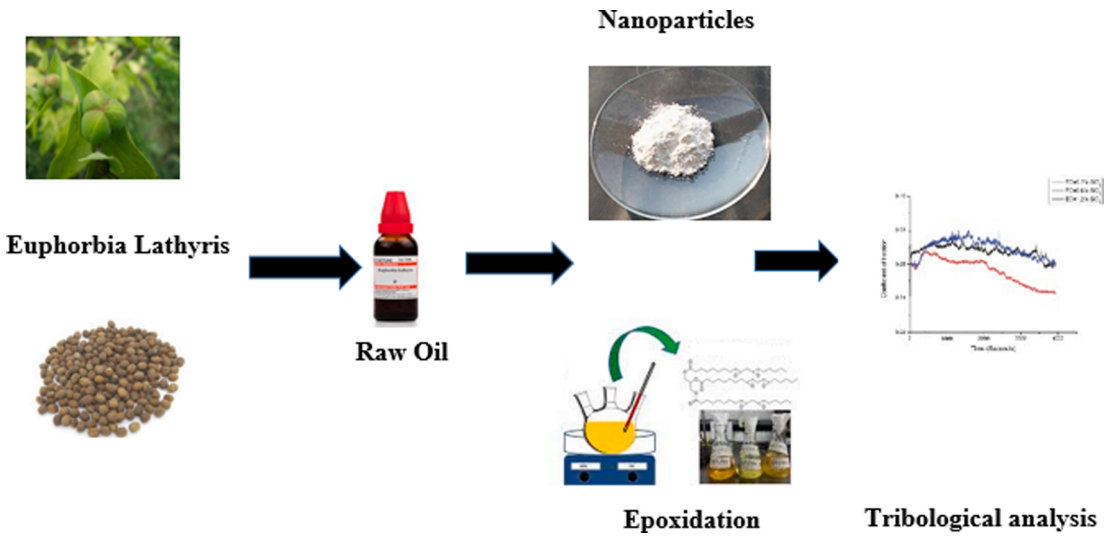


Fig. 3. Methodology adopted during this study.

The total yield of free fatty acid was determined by off-line GC-MS using a modified EN-14103 procedure as reported in the literature by Eq. (1).

$$\text{Total Fatty acid methyl esters} = \frac{(\sum A) - A_{IS}}{A_{IS}} * \frac{C_{IS} * V_{IS}}{W} * 100 \quad (1)$$

where $\sum A$ = total peak area of methyl ester, A_{IS} = peak area of methyl heptadecanoate C_{IS} = methyl heptadecanoate concentration in mg/ml, V_{IS} = used volume of methyl heptadecanoate solution in ml, and W = sample mass in mg.

The raw oil consists of maximum amount of oleic acid (77.23%) which was responsible for the wear of the surfaces as it is unsaturated fatty acid. The amount of saturated fatty acids are less in amount so, to make this oil feasible for lubrication purpose, chemical modification is required.

2.2. Chemical modification of the Euphorbia Lathyris oil

The chemical modification was performed to improve the thermo-physical properties of the oil. In the chemical modification process, the epoxidation process was performed. To perform this process, acetic acid was mixed with hydrogen peroxide in equal proportions while considering sulphuric acid as a catalyst for obtaining peracetic acid. Around 10.72 ml of *Euphorbia Lathyris* oil was mixed with 37.12 ml of peracetic acid. The mixture was heated at 20 °C for about 4.5 h so that complete reaction occurs. The obtained solution was emptied into the water which results in the rise of the epoxidized oil to the top surface. After leaving the mixture for about 1 h, water was extracted from the lower surface of the burette. The remaining oil was mixed with the ether to completely remove the water. The ether gets evaporated after continuous heating for about 4 to 5 h and the final solution was obtained.

For improvement of the lubricity characteristics of the vegetable oil, nanoparticles are significant enough. In boundary lubrication, nanoparticles help in building a defensive layer between the moving surfaces and also removes asperities from the surfaces. The zinc oxide nanoparticles are considered during this investigation as they form a polishing effect and disperse in the solution in a proper way [16]. The nanoparticles are prepared by a two step method and they are suspended in the epoxidized oil on a weight percentage basis (0.2, 0.5 and 0.8%). The surface modifier (Triton X-100, M/s Triveni chemical, Gujarat) was used to provide stability to the nanoparticles with the epoxidized oil. The amount of Triton X-100 was 50% (by weight) of each nanoparticle. For attaining steady suspension of the nanoparticles, they are mixed using a magnetic stirrer with 750 rpm/min for 3.5 h duration. A precise digital scale was used for weighing purpose of the nanoparticles and Triton X-100. Then the mixture was agitated for 45 min through the ultrasonic probe sonicator (M/s Samarth Electronics, Thane). The crystal structure of the nanoparticles was examined using X-ray diffractometer (M/s GNR India, Mumbai). XRD patterns were recorded using Cu- Kα radiation (λ = 1.5406 Å) at 40 kV and 30 mA from 7° to 80° (2θ) at a rate of 3° min^{-1}. The crystallite size of ZnO was calculated based on each XRD pattern, using Debye–Scherrer's equation [25]:

$$D = \frac{0.9 * \lambda}{\beta \cos\theta} \quad (2)$$

where D, λ, β and θ are the crystallites size (nm), the radiation wavelength, the full width at half of the maximum of the peak (radians) and the angular position of the peak.

2.3. SEM/TEm

The morphology of the nanoparticles were characterized with the help of Field Emission-Scanning Electron Microscope (FE-SEM 200 Quanta FEG) at 20 kV acceleration voltage. For the examination of worn surfaces of the pin, the surface was polished and keller's reagent was used for getting a clear examination of the worn surfaces.

Size of nanoparticles were analyzed with Transmission Electron Microscopy (TEM) (JEM-1400 Plus, JEOL Ltd.) with 120 kV acceleration voltage. Image J 1.50 was used for particle analysis by considering the procedure discussed in studies performed by Kaasalainen et al. [26]. Originally, 8-bit grayscale image was turned to a binary scale by adjusting the threshold with the "Threshold" tool. The threshold selection method was fixed inside one particle batch as the same thresholding method was impossible to use to all samples, due to the contrast differences arising from the size and density of nanoparticles. Particles were selected with "analyze particles" tool. In the case of ZnO nanoparticles, the circularity measure was used in order to exclude agglomerates from the analysis.

2.4. Dispersion analysis

The amount of dispersion of the nanoparticles depends on the concentration of it added to the base oil. The analysis of the dispersion of nanoparticles was performed by Ultra-Violet Visible Spectrometer procured from M/s Labtronics, Panchkula (LT-2201) with a wavelength range from 190 nm to 1100 nm having an accuracy of ± 0.3 nm. During different time intervals, the absorbance of the visible light passing through the samples was analyzed using visible spectrometry.

2.5. Zeta potential

Zeta potential is a physical characteristic that is shown by particles in suspension, macromolecules, or substance surfaces; it corresponds to the nanoparticle's electrical potential, as influenced by the nanocapsule ingredients and the medium in which they are distributed. This parameter is widely applied to indicate suspension stability in colloidal dispersions, where zeta potential values higher than 30 mV and lower than −30 mV promote high stability and prevent particles aggregation [27]. In this study, Zetasizer Ultra equipment having particle size range 0.3 nm to 10 μm was used.

2.6. Physicochemical properties

The kinematic viscosity was evaluated through the viscometer (M/s Swastik systems and services, New Delhi) based on the ASTM D-445 standard [28]. The viscosity was measured at atmospheric pressure and various temperatures (40 °C, 60 °C, 80 °C and 100 °C). The test was conducted three times and the mean value was mentioned to reduce the error and maintain reliability.

In order to calculate the viscosity index of the samples, the following equation was utilized according to the ASTM D-2270 [29]:

$$VI = \frac{[L - U]}{[L - H]} * 100 \quad (3)$$

where U represents kinematic viscosity at 40 °C, L and H are the kinematic visocisty at 40 °C with viscosity index 0 and 100, respectively, which have the same kinematic viscosity at 100 °C as the oil whose viscosity index are tried to be ascertained.

Flash point and respectively, using proper apparatus. Flash point was measured according to ASTM D92 [1]using a closed cup flash point tester (Pensky-Martens PMA 4). The gap of 1 °C is provided during testing with the start of 17 °C temperature. The lubricant samples were poured into the cup, heated and the cup was opened periodically to start the ignition. The pour point of the oil was measured according to ASTM D-97 [30]. The gap of 3 °C was maintained for the examination of the pour test during heating and cooling. The size of the sample was 20 ml for both the test. The acid value, iodine value, and density were measured based on the standard methods published earlier [31]. Table 1 shows the values obtained for ZnO nanoparticle and *Euphorbia Lathyris* oil.

Table 1
Properties of the ZnO nanoparticle and *Euphorbia Lathyris* oil.

Materials Properties Nanoparticle	Purity (%)	Size (nm)	Density (kg/m^3)	Color	Shape
ZnO	99.9	40	5812	White	Spherical
Properties	***Euphorbia Lathyris* oil value**			**Epoxidized Oil**	
Kinematic viscosity (cSt) at 40 °C	42.13 ± 2.09			141.19 ± 2.37	
Kinematic viscosity (cSt) at 100 °C	8.12 ± 0.29			12.93 ± 1.28	
Viscosity Index	147			187	
Density (kg/m^3) at 15 °C	917.03 ± 1.03			963.12 ± 2.18	
Flash point (°C)	78.12 ± 2.14			159 ± 2.42	
Pour point (°C)	−3.12 ± 0.37			2.14 ± 0.83	
Iodine value	89 ± 2.56			17.12 ± 1.04	
Acid value (mg KOH/g)	37.12 ± 0.21			19.12 ± 1.02	

2.7. FTIR spectroscopy

The Fourier Transform Infrared Spectroscopy (FTIR) was used for the characterization of the Epoxidized oil. The Shimadzu FTIR 8400 s machine equipped with MIRacle TM ATR was used to obtain an infrared spectrum of absorbed emissions. The range of the wavelength during the examination was 400 to 4000 cm^{-1} with a resolution of 4 cm^{-1}. FTIR was used to determine the change in functional groups that contribute to the synthesis of each system, as well as to verify the number of epoxide groups.

2.8. Material for pin and disc

The LM 13 alloy (copper: 1.3%; Manganese: 0.4%; Tin: 0.1%; Titanium: 0.2%; Manganese: 1.2%; Silicon: 11.4%; Iron: 1.0%; Nickel: 1.4%; Zinc: 0.5%; and Lead: 0.1%) was procured from the (M/s Samarth metal corporation, Mumbai) which was used as the pin material for the investigation. The purpose of consideration of this material due to its application for the piston which faces maximum friction and the hardness is 98 HRB. This type of alloy is also capable to resist wear and corrosion. For the Disc, EN 31 steel was used as it contains more amount of hardness (62 HRC) and it is highly worn resistive. After purchasing ingots from the supplier, the pin with proper dimensions was made using turning operation on the Lathe machine. The pin was further cylindrically shaped. The one end of the pin was made spherical for getting the point effect while having contact with the disc. The pin consists of 10 mm diameter and length was 30 mm. The pin was further polished by using emery paper of following grit sizes 200, 400 600 and 1200 nm before performing an experiment on the machine.

2.9. Experimental set up

The pin on disc machine was purchased from the M/s DUCOM, Bangalore, India that was used for investigating the friction and wear characteristics during the application of lubricants. Fig. 5 shows the schematic image of the mechanism involing point contact of pin and disc. The friction force was obtained on the screen with the help of the load cell attached to the tribo machine. For obtaining the friction coefficient, friction force was normalized by the load applied. The lubricant was provided in a drop wise way to the interfaces with the help of the pump operated by the electric motor. The scanning electron microscope (SEM) was used to study the worn out surfaces during the analysis. Table 2 mentioned the conditions considered for the examination of the samples.Fig. 6.

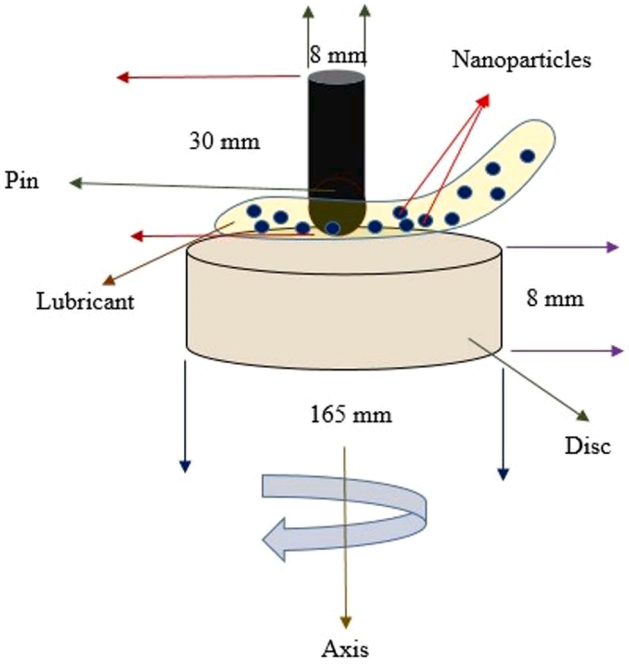


Fig. 5. Schematic image of the mechanism involved in DUCOM machine.

Table 2
Condition taken during the analysis.

Conditions considered	Value
Applied load (kgf)	10 ± 5
Slidng speed (rpm)	150 ± 50
Temperature (°C)	75 ± 3
Duration (s)	3600

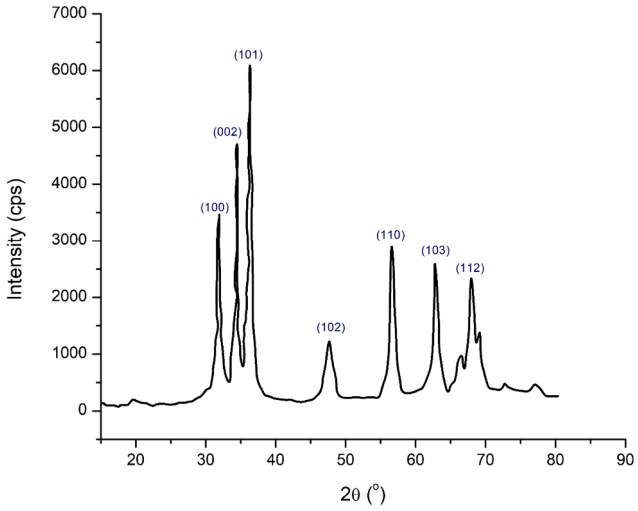


Fig. 6. XRD spectrum of the ZnO nanoparticles.

3. Results and discussion

3.1. XRd

The sharp diffraction peaks of the ZnO nanoparticles indicate their good crystallinity. The diffraction peaks located at 31.72°, 34.36°, 36.27°, 47.42°, 56.38°, 63.18° and 68.12° correspond to the (100), (002), (101), (102), (110), (103), and (112) reflection planes of

hexagonal structure of ZnO (JCPDS: 03-065-3411), respectively. Characteristic peaks related to impurities were not detected in the XRD pattern confirming the high purity of the sample. The calculated unit cell parameters for the ZnO nanoparticles are a = b = 3.2762 Å and c = 5.23481 Å.

3.2. FTIR analysis

Fig. 7 shows the FTIR spectrum evaluated for the raw oil and epoxidized *Euphorbia Lathyris* oil. The C = C peak got disappeared in the epoxidized FTIR spectrum clearly indicating its consumption. Due to the epoxidation process, evidence of the bond C-O-C was formed at 821.23 cm^{-1} which attributed to the epoxide group. Similar kinds of results were also reported in the FTIR spectrum of epoxidized olive oil in the literature [32]. The presence of a hydroxyl group (-OH-) in the epoxidized oil was observed in the range 3240–3530 cm^{-1}. The result of no epoxide peaks explains the side reactions and oxirane cleavage occurred that degrades the epoxide rings [2,33].

3.3. Fatty acid and epoxidized oil characterization

Fig. 8 shows the amount of fatty acids present in the *Euphorbia Lathyris* oil. The most dominant factor in the raw *Euphorbia Lathyris* oil is the presence of oleic acid contributing to around 80%. The presence of the oleic acid helps in resisting the applied pressure on the surface during their contact which maintaining a proper film on the asperities. Around 2.8% amount of linoleic acid is also present in the raw oil. The least composition is of palmitoleic acid which doesn't contribute much. The *Euphorbia Lathyris* oil also possesses cold flow behavior which promotes it to consider for the lubrication applications.

Fig. 8 and Table 3 shows the major compositions of epoxidized fatty acid methyl esters (EFAME). Small amount of unsaturated fatty acids were found in EFAME after epoxidation. A small amount of compounds containing a hydroxyl group (No.9, 26.316 min; and No. B, 24.531 min) were detected, which were the by-products of epoxidation. It was remarkable that both compounds 8 and 10 contains one and two hydroxyl groups, respectively, were by-products of epoxidation, while compound 8 readily reacted with carboxylic acid, and the main reason is that the hydroxyl group of compound 8 is a primary hydroxyl group with high activity.

3.4. Morphology and size analysis

Scanning electron microscope (SEM) studies were carried out to find out the surface morphology of zinc oxide nanoparticles (NPs). SEM

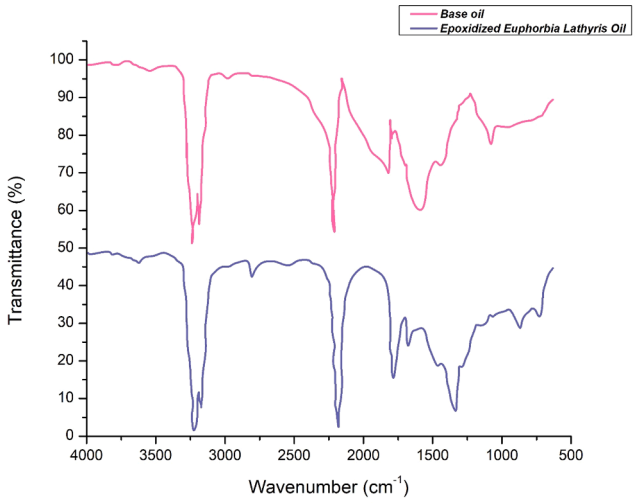


Fig. 7. FTIR spectrum of the epoxidized oil and raw *Euphorbia Lathyris* oil.

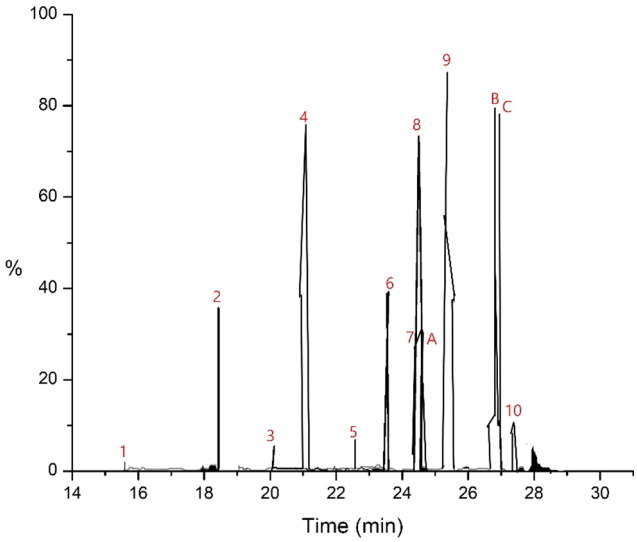


Fig. 8. GC–MS chromatogram of epoxidized oil.

Table 3
Expoxidized oil peak with retention time.

Peak No.	Retention time	Area (%)
1	15.127	0.78
2	18.321	5.17
3	20.719	0.81
4	21.391	43.12
5	22.078	1.29
6	23.814	14.82
7	24.182	5.2
8	24.871	0.27
9	25.871	2.28
10	27.814	1.78
A	25.021	1.83
B	26.921	11.31
C	27.031	10.12

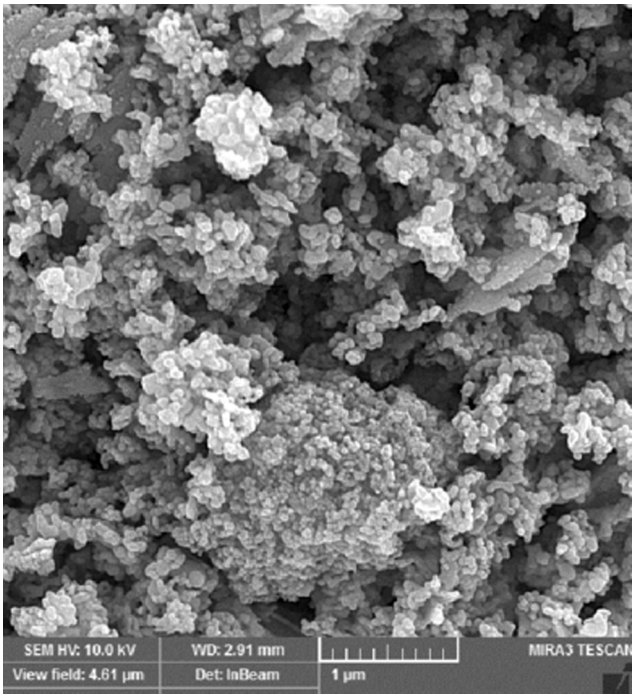
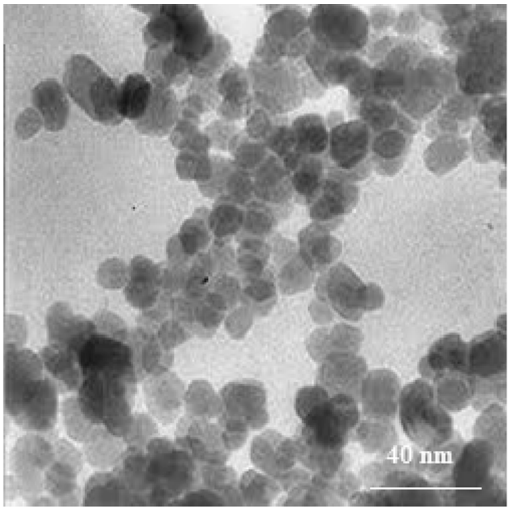


Fig. 9. SEM image of ZnO NPs.

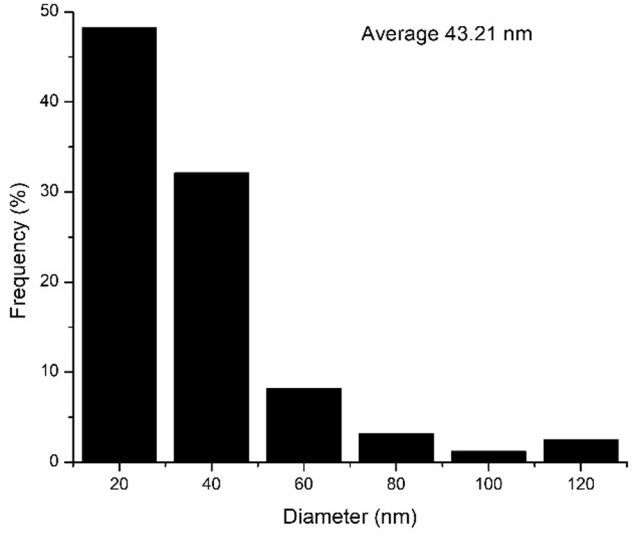
micrographs of the zinc oxide NPs have been represented in Fig. 9. SEM studies show zinc oxide is in pure form and particles are beautifully white colored nanoparticles. Transmission electron microscopy (TEM) studies were carried out to find out the exact particle size of ZnO NPs. Fig. 10 shows the TEM image of the ZnO nanoparticles. TEM images show that ZnO nanoparticles are having particle size in the range of 38 nm – 46 nm [Fig. 10 (a)]. The size distribution histogram for nanoparticles provided their respective size as 43.12 ± 16.9 nm [Fig. 10 (b)].

3.5. Dispersion stability of nanoparticles

The dispersion stability of the nanoparticles depends upon the soluble capacity of the nanoparticles. For examining the dispersion stability, UV spectroscopy dispersion analysis was conducted at 450 nm wavelength. Fig. 11 shows the dispersion stability of the ZnO nanoparticles to the modified oil. Before considering the investigation, the nano lubricant samples were placed in an ultrasonic bath at 25 °C temperature for 45 min. The analysis was performed for 120 h. It was observed that there were no major changes appeared in the solution with



(a)



(b)

Fig. 10. (a) TEM images of ZnO nanoparticles (b) Frequency percentage of nanoparticle size diameter.

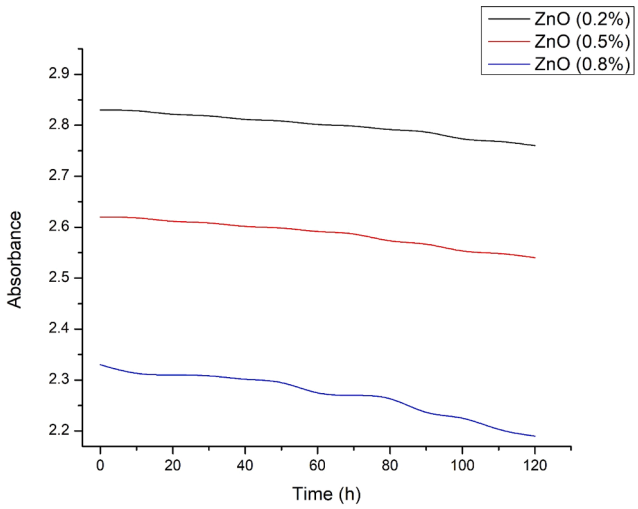


Fig. 11. Dispersion stability of the chemically modified oil with nanoparticles.

chemically modified oil and nanoparticles concentration of 0.2%, 0.5%. This shows that ZnO nanoparticles were properly dispersed into the modified oil. There are some of the changes that occurred during 0.8% nanoparticles in addition to the chemically modified oil. There was precipitation occurred in the solution and agglomeration of nanoparticles increases their density which promotes their separation from the fluid [34].

The agglomerating between nanoparticles was characterized by the Zeta potential value. The bigger absolute value means the nanoparticles disperse more homogenously. The absolute value of ZnO Zeta potential with different concentrations was shown in Fig. 12. As shown in Fig. 12, Zeta potential absolute value of ZnO (0.5%) nanoparticles were greater than that of ZnO (0.8%) nanoparticles. This indicated that static repellency of ZnO with 0.5% become stronger than the ZnO with 0.8% concentration. The ZnO with 0.2% and 0.5% was well dispersed in lubricating oil.

3.6. ZnO nanoparticles characterization

3.6.1. Viscosity increment

Fig. 13 shows the viscosity increment in percentage when the ZnO nanoparticles are added to the epoxidized oil in different proportions. It can be seen that the addition of the nanoparticles increases the viscosity with comparison to the epoxidized oil. The viscosity of all the lubricants

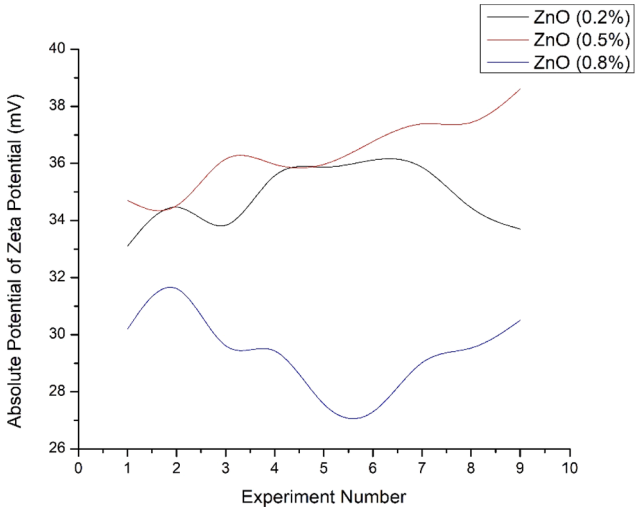


Fig. 12. The absolute value ZnO nanoparticles zeta potential.

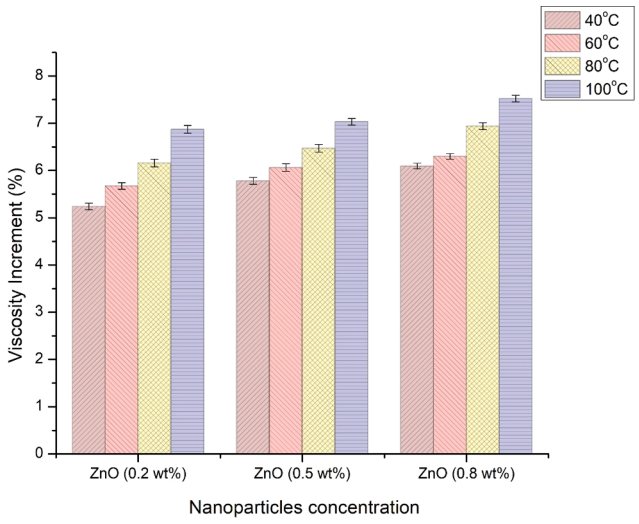


Fig. 13. Viscosity increment measurement for different ZnO nanoparticles concentration.

decreases with an increase in temperature. The maximum increment in viscosity was found at 0.8% and 100 °C. The increase of nanoparticles results in agglomeration on the surface and also collision increases which result in an increase of nano lubricant viscosity. The collision of the molecular forces with increase in temperature reduces the viscosity of the nano lubricants [25]. The mathematical relation for the viscosity increment measurement is mentioned in Equation (4). Asnida *et al.* [16] also stated the viscosity increment with addition of the copper oxide nanoparticles.

$$Viscosity\ \ increment(\%) = \frac{Nano\ \ lubricant\ \ viscosity - Epoxidized\ \ oil\ \ viscosity}{Epoxidized\ \ oil\ \ viscosity} * 100 \quad (4)$$

3.6.2. Relative viscosity

The relative viscosity with change in temperature for different concentrations is shown in Fig. 14. The maximum change can be observed for the 0.8% ZnO addition at maximum temperature. The reason for the rise is already discussed in the previous sub-section. Eq. (5) shows the mathematical relation for getting relative viscosity. The change in viscosity was attained during an increase in temperature due to the movement of the molecules in the fluid. The same reason has also been stated by Kerni *et al.* [32].

$$\mu_r = \frac{\mu_{nl}}{\mu_{eo}} \quad (5)$$

where, μ_r = relative viscosity; μ_{nl} = viscosity of nano lubricant; and μ_{eo} = viscosity of epoxidized oil.

3.6.3. Viscosity index increment

Viscosity index is another important parameter that decides the lubricity and it is based on the kinematic viscosity of the samples. Fig. 15 shows the variation of viscosity index increment with respect to the temperature for various concentrations of the nanoparticles. The maximum viscosity during the addition of the nanoparticles was obtained at 0.8% and maximum viscosity was obtained at a temperature 100 °C. With an increase in kinematic viscosity of the samples, viscosity was increased at a temperature 40 °C and 100 °C.

3.6.4. Flash and pour point

The flash point is the lowest temperature at which the fluid is capable to form a flammable mixture with air and the pour is the lowest temperature at which it starts flowing. Fig. 16 (a) and (b) shows the effect of ZnO nanoparticles on the flash point and pour point respectively. With an increase in the concentration of the nanoparticles, an increase in the value of flash point was observed which was a good indication. The increase in flash point provides an increased operating limit of the temperature of the lubricant. Due to an increase in the limit of operating temperature, more stability, and better thermophysical properties of ZnO nanoparticles were observed. The better flash point were also observed with the addition of nanoparticles in the work reported by Battez *et al.* [35]. Hence, the enhanced flash point can be considered as

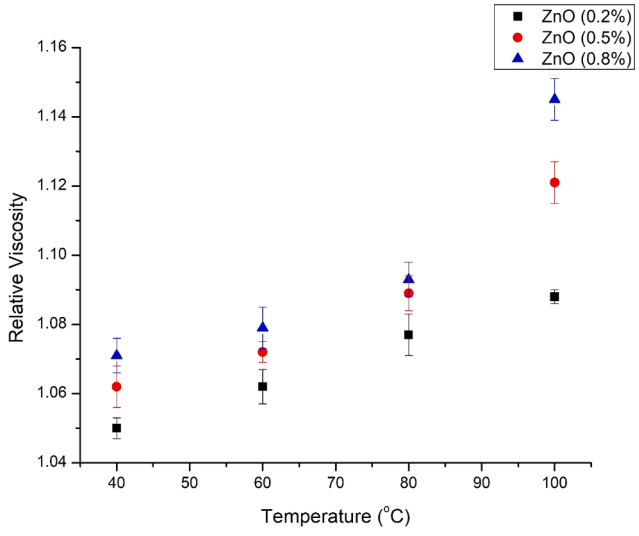


Fig. 14. Variation of relative viscosity with temperature.

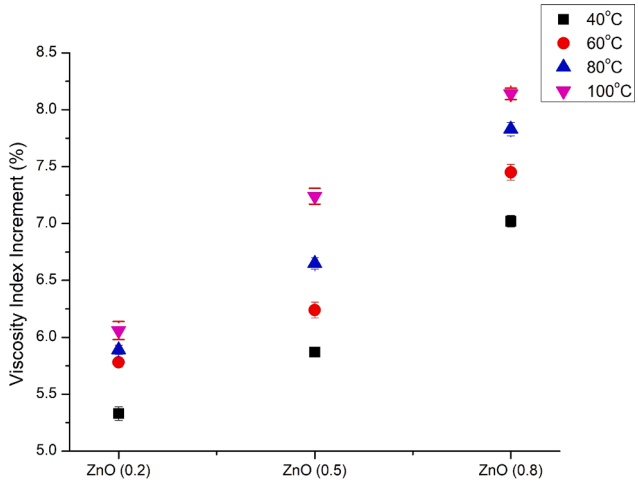


Fig. 15. Viscosity index increment with respect to the nano lubricants at various temperatures.

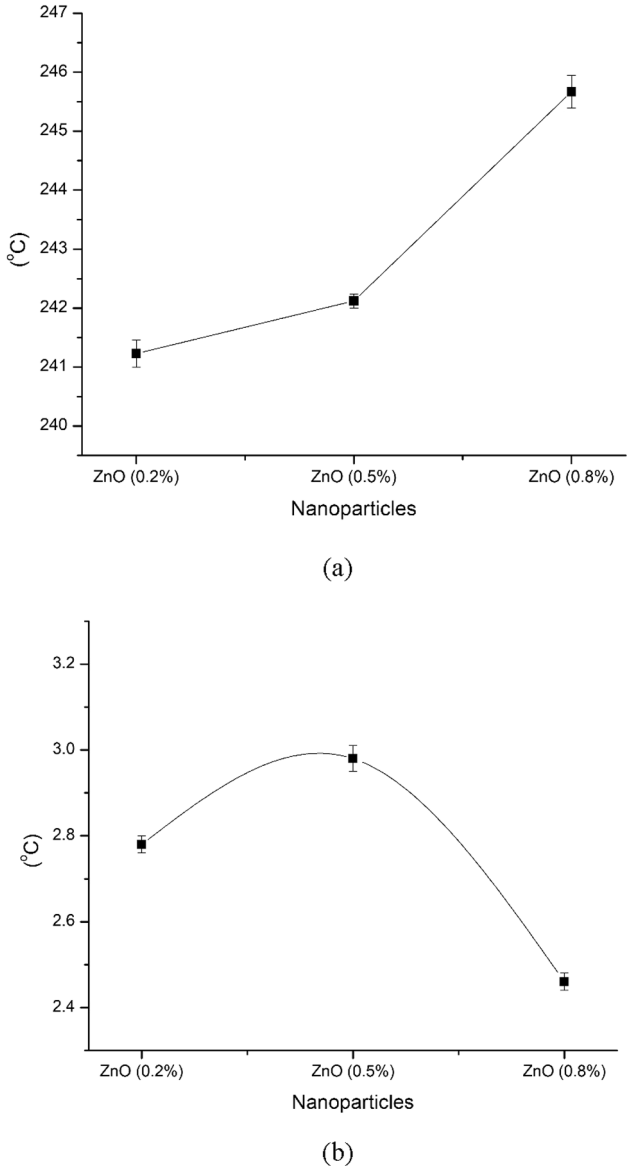


Fig. 16. Effect of nanoparticles addition on the (a) flash point (b) pour point.

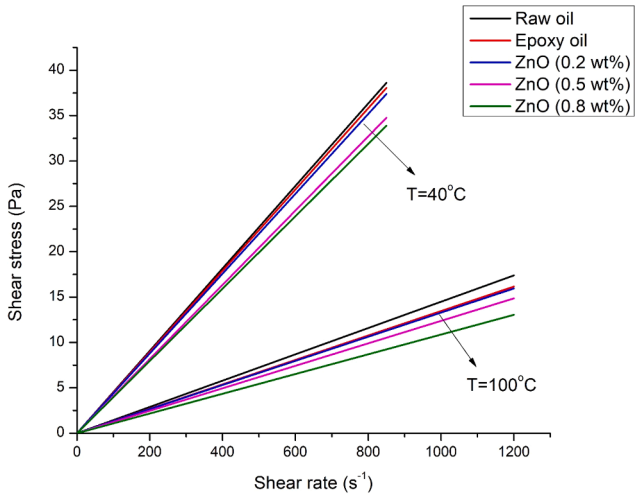


Fig. 17. Shear rate with variation of shear stress for different lubricants.

an advantage in relation to enhanced lubricity characteristics of epoxidized oil. With the addition of the nanoparticles, the pour point decreases at 0.8%. This was due to the increase in the thermophysical properties of the nano lubricants. The increase in temperature resists the flow of the lubricant and a higher concentration of the nanoparticles doesn't provide better flow due to their agglomeration on the surface as reported in the literature [6,16,19,36].

3.7. Rheology

Fig. 17 shows the shear stress variation with the shear rate for all lubricant samples. For both the temperature, the trend comes out linear which confirms the Newtonian characteristics of the samples considered for the study. The change in viscosity of the samples with respect to the shear rate is depicted in Fig. 18. With an increase in temperature, there is more shear stress with a reduction in viscosity. At 40 °C temperature, there was not many changes occurred in the value of shear stress as compared to 100 °C temperature. The epoxidized oil doesn't show more variation in the viscosity of the fluid with respect to temperature due to a more complex saturated structure. With an addition of the nanoparticles to the epoxidized oil, more resistance to the shear rate was obtained with a limit to 0.6%. During the addition of 1.2% nanoparticles, more shear stress was achieved due to the agglomeration of nanoparticles at the lubricant film. There are negligible changes observed in the viscosity of the tested samples which confirms their Newtonian behaviour. The same trend has also been observed in the study done by Kerni *et al.* [32].

3.8. Coefficient of friction characterization

Fig. 19 shows the friction behavior of the raw oil and the chemically modified oil. It has been observed that epoxidized *Euphorbia Lathyris* oil presented a minimum friction coefficient with respect to the raw oil as shown in Fig. 19 (a). This was due to the formation of a thick film that was created by the tribochemical reaction. It has been reported that the formation of the oxirane ring provides an effective layer on the surfaces during the motion. The raw oil initially shows the minimum coefficient of friction which showed increased behavior after 1670 s due to the instability of the film formed on the surfaces.

The effect of nanoparticles addition is depicted in Fig. 19 (b). The minimum coefficient of friction was obtained when the concentration of 0.5% is there in the chemically modified oil. The coefficient of friction was reduced by about 33.12% when compared to the raw oil. The mechanism of friction reduction involves the filling of gaps on the surfaces with nanoparticles which provides smoother and better lubricity. With the addition of the nanoparticles, contact surface area increased

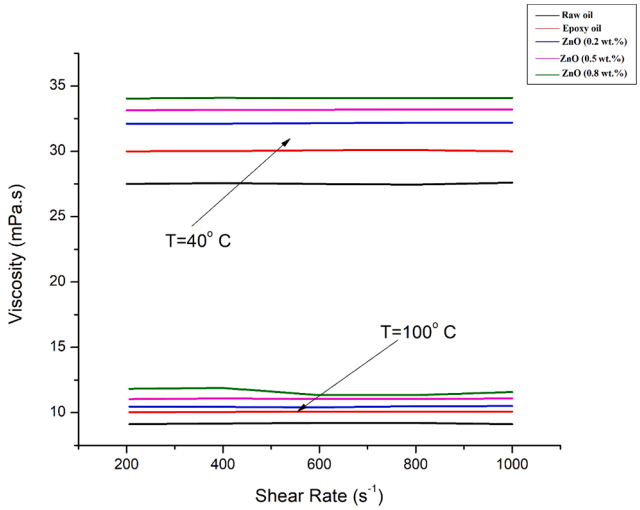


Fig. 18. Viscosity with respect to shear rate for different lubricant samples.

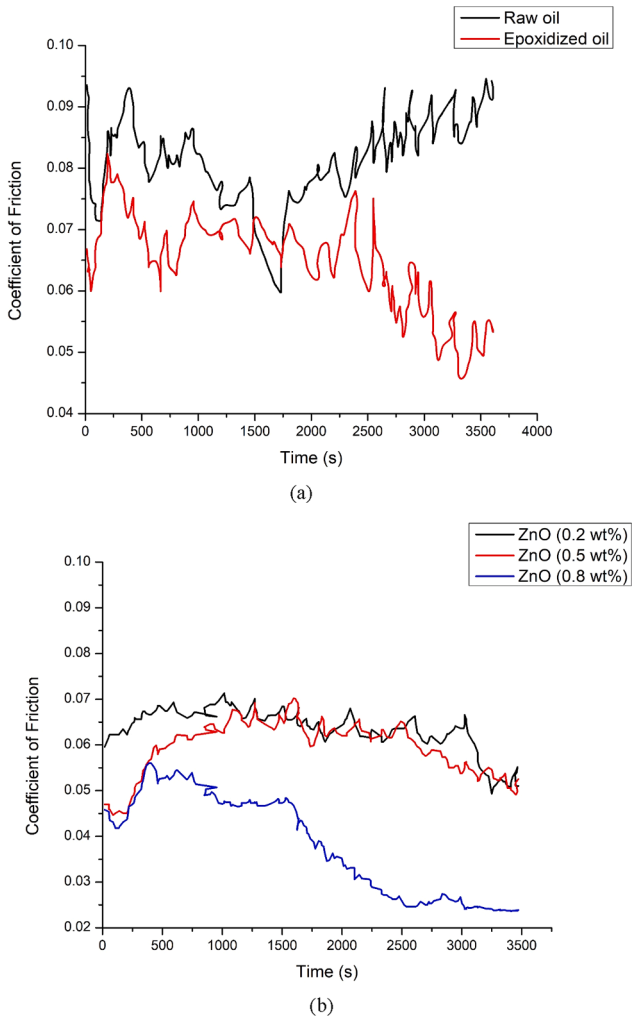


Fig. 19. Coefficient of friction (a) raw *Euphorbia Lathyris* oil (b) nano lubricants.

which resulted in the reduction of the applied pressure and rolling mechanism occurs instead of sliding friction. The same reason has also been reported by Ghaednia *et al.* [37].

Based on the above results, it can be inferred that an optimum concentration of the nanoparticles is desired to received better tribomechanism. When the concentration becomes lower e.g. 0.02% and 0.03% etc., there is a limited conversion of the sliding motion into rolling one which attributed more friction. With an increase in the amount of nanoparticles after a limit, particles get aggregated on the metals which enhance friction and wear on the surface during their contact. Fig. 20 shows the tribomechanism involved while considering the proportions of the nanoparticles. It can be observed that there is a requirement of optimum concentration of the nanoparticles to maintain proper lubricant film on the surface. Higher concentration in the nanoparticles produces abrasive behavior which results in a reduction of the nanoparticles due to resistance to the flow of the lubricant. The same hypothesis has also been reported in the previous studies and also presenting abrasive wear with increased nanoparticles concentration [16,38,39].

3.9. Wear

The wear of the pin was minimum for the epoxidized *Euphorbia Lathyris* oil with comparison to the base oil as evident from Fig. 21. The

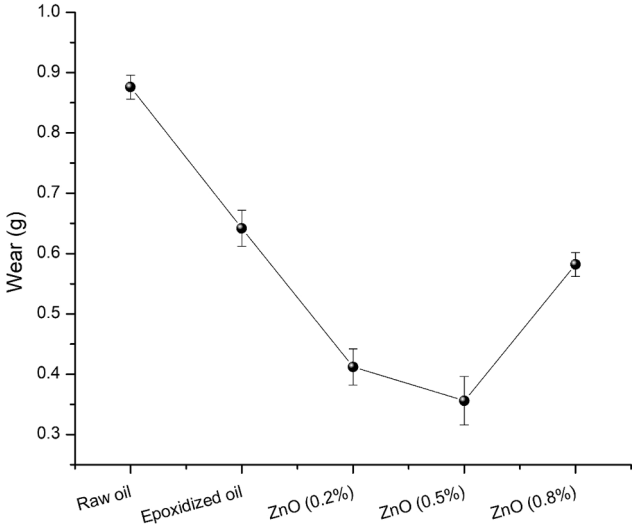


Fig. 21. Effect of nanoparticles on the wear.

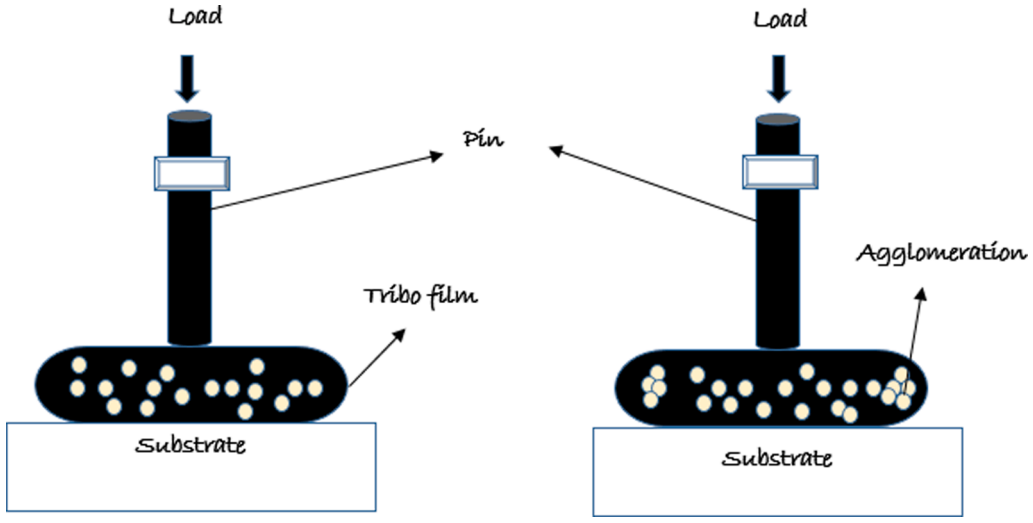


Fig. 20. Schematic description of the mechanism (a) low agglomeration of the nanopartciles and better lubricity (b) high agglomeration of the nanopartciles and wear of the surface.

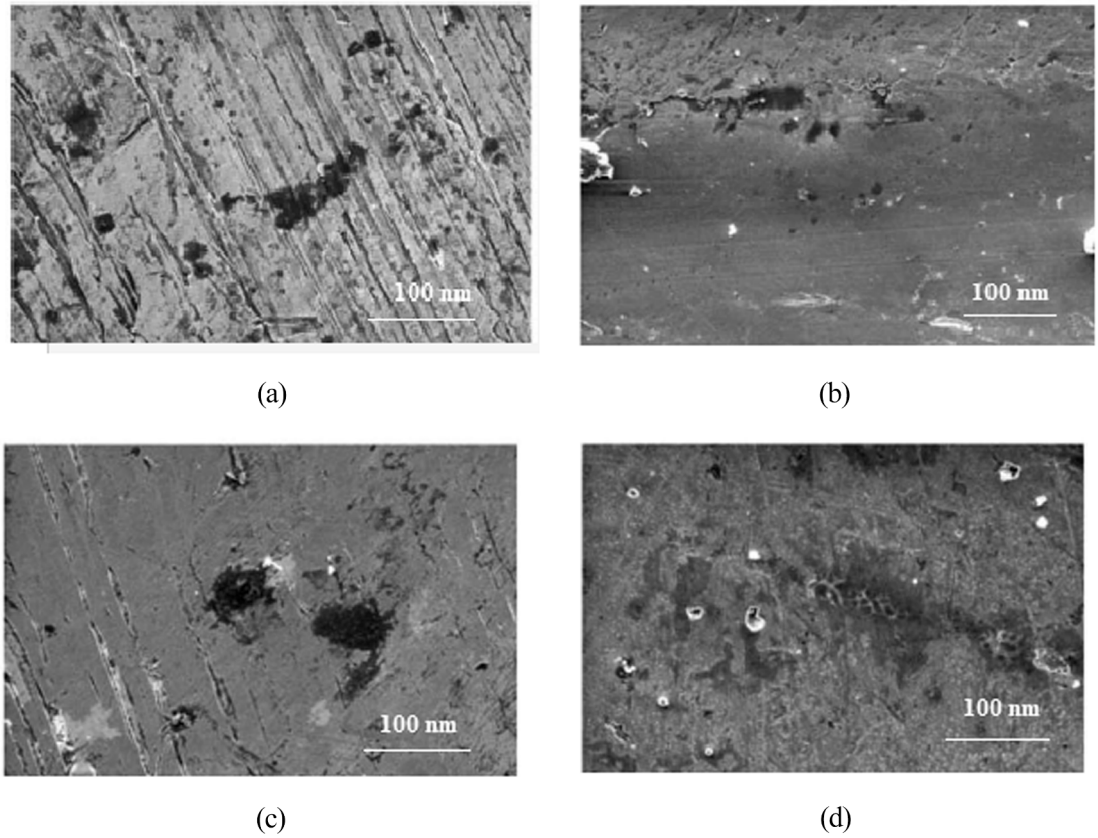


Fig. 22. SEM images of the lubricants for (a) Raw *Euphorbia Lathyris* oil (b) Epoxidized *Euphorbia Lathyris* oil (c) ZnO (0.5%) (d) ZnO (0.8%).

epoxidized oil forms an improved lubricant layer between the surfaces during their contact. The epoxidized oil contains polar groups which promote adsorption on the surface resulting in minimum wear.

The concentration of ZnO nanoparticles up to 0.5% also showed minimum wear of the pin when compared to the raw *Euphorbia Lathyris* oil. The reduction in the wear was around 8.24%. The nanoparticles effect on the anti-wear property depends on the proper dispersion of the particles in the lubricant. The ZnO nanoparticles are having the better capability to gets properly dispersed in the solution which promotes anti-wear mechanism.

3.10. Worn surface analysis

Fig. 22 shows the SEM images of the lubricant samples obtained for the worn surface study. Fig. 22 (a) shows the SEM images for the base oil. More scratches on the surface have been observed and the occurrence of the ploughing effect results in the formation of these scratches. For epoxidized *Euphorbia Lathyris* oil as depicted in Fig. 22 (b), the absence of the ploughing effect and the delamination of the surface was there which results in the formation of the smooth surface. This was due to the formation of -O- cross-linking on the surface which protects it by assisting the lubricant film. The same reason for the epoxidation of the lubricants are reported in the literature [2,40,41]. The addition of the nanoparticles to the chemically modified oil plays an effective role by providing a protective layer on the surface. The addition of the nanoparticles up to 0.5% results in less wear as they provide a defensive film between the surfaces during their motion. Fig. 22 (c) shows SEM images with the effect of 0.5% copper oxide nanoparticles addition. The minimum damage to the surface was observed which was due to the proper filling of the asperities present on the surfaces and provide resistance to the applied pressure. Nano-particles in the lubrication oil can fill the scar and grooves on the friction surface. The ZnO nanoparticles added to the lubricant and provide a rolling mechanism instead of the mending effect which reduces wear between the surfaces during the sliding motion [38,42].

However, a further increase in nanoparticles results in more amount of wear due to adhesive nature. The adhesive wear occurs when there is a removal of material due to adhesion between two surfaces, and there are two types of adhesive, which are mild adhesion and severe adhesion. Mild adhesion occurs at a low rate and it removes surface films such as oxides. While, severe adhesion removes the metal due to tearing, breaking and melting of metallic junctions. Fig. 22 (d) shows the effect of adhesion and abrasion on the surface when amount of copper oxide nanoparticles further increased. This was due to the agglomeration of the nanoparticles on the surface and results in the wear of the surface [19]. The anti-wear mechanism of nano-particles additive is as follows: when the lubricant film between two pairs becomes thinner and mixed lubrication or boundary lubrication occurs, the nano-particles may carry a proportionate load and separate the two surfaces to avoid from adhesion. Therefore, it is beneficial for anti-wear properties. The undesirable removal of solid during sliding or rolling motion called wear. Wear analysis is necessary when investigating worn between two or more surfaces to determine the type of wear occurred.

4. Conclusions

The *Euphorbia Lathyris* oil was chemically modified and the effect of nanoparticles at various concentrations were evaluated for the tribological applications. Based on the observations, an improved behavior by the application of nanoparticles to the epoxidized oil is achieved. In the study of the kinematic behavior of the nano lubricants with respect to the nanoparticles and increase in temperature from 40 °C to 100 °C , an improvement in the viscosity was observed during the addition of the ZnO nanoparticles. With the addition of the nanoparticles, an increase in the viscosity index of 3.17% was observed. A higher amount of increment of 4.82% was obtained at higher concentration. The addition of the

nanoparticles to the chemically modified oil increases the flash point around of the nano lubricants from 247 °C to 254 °C . A higher value of the flash point was obtained at 0.8% which was around 5.12% increment from the epoxy oil. Adding nanoparticles to the fluid decreases the pour point at 0.8% concentration. There are marginal changes in the pour point at lower nanopartciles concentration i.e. 0.2% and 0.5%. From the flow behaviour, it has been observed that the lubricants considered for the test follow Newtonian behaviour as there is a linear relationship between shear rate and shear stress which confirms it. Better stability of the nanoparticles were observed up to 0.5% addition. The epoxidized oil shows improved lubricity in comparison to the raw *Euphorbia Lathyris* oil in terms of reducing friction. This was due to the presence of -O- cross linking in the surfaces which successfully forms a protective film on the surface. The addition of the ZnO nanoparticles to the chemically modified oil improves the lubricity of the samples. With the addition of up to 0.5% concentration, better results are obtained in terms of reducing friction. Their addition to the lubricant assists in forming a better defensive film between the surfaces and prevents direct contact of metals during the motion. The raw oil shows maximum damage of the surface with comparison to other lubricants. This was due to the delamination of the surface. The epoxidized oil shows less damage of the surface with comparison to the raw oil as there is presence of double bonds in the molecular structure. The addition of the nanoparticles up to 0.5% to the modified oil shows less wear and damage to the surface. This was due to proper dispersion of the ZnO nanoparticles on the surface. The *Euphorbia Lathyris* oil with ZnO nanoparticles as additives are having advantages for tribological applications. The concentration of ZnO nanoparticles upto 0.5% provides better anti-wear characteristics and the dipersion of nanoparticles to the oil was also found stable. This type of lubricants are suitable for the application to automotive sector.

CRediT authorship contribution statement

Yashvir Singh: Writing - original draft, Conceptualization. **Nishant Kumar Singh:** Writing - original draft. **Abhishek Sharma:** Methodology. **Amneesh Singla:** Writing - original draft. **Deepak Singh:** Writing - original draft. **Erween Abd Rahim:** Writing - original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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